

Taking a whole-of-system approach to food packaging reduction

Sabrina Chakori^{a,*}, Russell Richards^{b,c}, Carl Smith^b, Nicholas J. Hudson^a, Ammar Abdul Aziz^a

^a School of Agriculture and Food Sciences, The University of Queensland, Brisbane, Queensland, 4072, Australia

^b School of Business, The University of Queensland, Brisbane, Queensland, 4072, Australia

^c Coastal and Marine Research Centre, Griffith University, Gold Coast Campus, QLD, 4222, Australia

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ABSTRACT

Food packaging use is a wicked problem that is increasingly impacting on the environment. Current efforts to solve this global challenge are fragmented, fixating on single issues rather than on a 'whole-of-system' approach. To address this, we have used a novel approach combining systems thinking and network analysis to map out food packaging use in food systems. This process draws from experts across the Australian food system. We report three main results. First, the resulting Causal Loop Diagram sheds light on the experts' mental models about food packaging use. Second, current interventions fail because they tackle the symptoms of the problem and not the drivers. Third, the ten most influential food packaging drivers are presented and belong to the globalisation and household subsystems. The findings highlight that dependence on packaging is a symptom of the growth-driven globalised food market and time-deprived society. Finally, we demonstrate that shifting the current food systems towards economic degrowth principles could potentially curb the use of food packaging. Under a degrowth framework the need for packaged food could be reduced by eliminating the tension between cooking and working time constraints. Designing for a not-for-profit food economy and re-localising it would also reduce the need for packaging.

1. Introduction

The reliance on packaging in food systems is a global challenge that can be classified as a 'wicked problem'. Wicked problems are characterised by complexity, due to multiple possible causes and internal dynamics that change over time (Rittel and Webber, 1973; Peters, 2017). The food market is responsible for up to 35% of global packaging production (Rundh, 2005). Packaged food is ubiquitous throughout the world, with the global packaged food market worth US\$2.4 trillion in 2014 (Clapp and Scrinis, 2017). Food packaging is used to protect food, facilitate food transport and storage, and for communication and marketing to consumers (Gustavo et al., 2018). The main material used for food packaging is fossil-fuel based plastics (Van Crevel and Khristolyubova, 2016). The use of packaging causes environmental degradation (Geyer et al., 2017; Otto et al., 2021; Scharpenberg et al., 2021; Sundqvist-Andberg and Åkerman, 2021). This includes greenhouse gas (GHG) emissions from plastic production, which account for 6% of total oil production (Van Crevel and Khristolyubova, 2016; Kaza et al., 2018), and pollution of land and marine environments (Jambeck et al., 2015) — 31% of plastic debris found in marine ecosystems originates from

food and beverage packages (Van Crevel and Khristolyubova, 2016). Additionally, food packaging can also result in detrimental impacts on human health. This could be through the exposure to food contact chemicals (e.g. bisphenol A, ortho-phthalates, perchlorate and poly-fluoroalkyl substances), which are hazardous substances found in food packaging materials that migrate into food, increasing the risks of adverse health effects, such as coronary artery disease, endocrine disruption or carcinogenicity (Srouf et al., 2019; Muncke et al., 2020).

Food packaging is at the centre of public and political attention in many countries (Sundqvist-Andberg and Åkerman, 2021). Many of the proposed solutions that address the wicked problem of food packaging focus on the packaging itself. This includes recycling (Coates and Getzler, 2020) and the development of alternative 'sustainable' packaging materials (Yano et al., 2014; Van Crevel and Khristolyubova, 2016). However, focusing solely on packaging only addresses the symptoms of the problem and does not explain the underlying systemic causes for the rapid growth in, and reliance on, food packaging in general. In fact, current solutions mainly target the disposability and health effects of primary packaging, which is the material in direct contact with the product and with consumers. However, food systems also include

* Corresponding author.

E-mail address: sabrina.chakori@uq.net.au (S. Chakori).

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secondary packaging, which provides collation and enables shelf loading, and tertiary packaging, which is designed to reduce the stress and damages along the supply chain (Inns, 2012). This highlights the need for applying a 'systems thinking' approach when analysing food systems (Editorial, 2020a, 2020b; Ingram et al., 2020). Interventions can only result in systemic change if informed by the system structure (Polhill et al., 2016). The design of sustainability policies should consider systems thinking so that they encapsulate the complex dynamics of interconnected variables and their characteristics (Honti et al., 2019).

Food systems are integral to the social, environmental and economic fabric of societies and encapsulate a multitude of actors and their complex interlinked activities (Nguyen, 2018). To tackle this complexity, this study draws on systems theory, which is a scientific exploration of 'wholeness', in which elements are interrelated with themselves and with the environment (Von Bertalanffy, 1972). This study investigates the system behaviour (Stermann, 2000; Hjorth and Bagheri, 2006) of the food system using systems thinking, complemented by a network analysis. The aim of this work is to understand the determinants and potential leverage points of the wicked problem of food packaging use. Understanding the systemic causes of the problem allows the rational design and implementation of solutions that could transform the food economy so that it is more sustainable and less reliant on food packaging. Hence, the questions are: how much do we understand the system driving food packaging? Is the lack of a full system's understanding a barrier to systemic change? This study explores Australian experts' knowledge on the interdependencies and feedback paths of the drivers of food packaging use, specifically investigating their understanding of the drivers of the use of food packaging and potential solutions proposed by them able to reduce the negative effects caused by it.

In the following section, we describe the methods (section 2), i.e. systems thinking, the interviews process and the network analysis. Section 3 presents the results of the expert interviews, the causal loop diagram, the network map and radiality analysis of the main food packaging use drivers. Section 4 provides a discussion on (4.1) the existing gap between food packaging drivers and current interventions and (4.2) introduces to economic degrowth as potential solution able to improve the sustainability of food systems. Finally, section 5 draws the conclusion and suggests possible future research.

2. Methods

2.1. Causal loop diagram development

This study expanded on the Causal Loop Diagram (CLD) developed by analysing the behaviour over time of various food packaging drivers since the 1960s, and by a narrative literature review of scholarly articles and organisational reports (Chakori et al., 2021). A CLD is a systems technique used to develop a dynamic hypothesis (Stermann, 2000), which is a qualitative representation of how components within a system interact to generate the observed system behaviours.

The produced CLD contains variables that represent the drivers of food packaging use. Variables are connected by arrows that indicate the direction of causality (cause → effect) and use polarities to indicate direction of correlation between the interconnected variables; i.e. '+' if the cause and effect move in the same direction and '-' if they move in opposite directions. A feedback loop is formed when a sequence of cause-and-effect connections return to the initial cause variable. Loop labels are used in the CLD to indicate and characterise the feedback loops into reinforcing (represented by an R) or balancing (represented by a B) (Sherwood, 2011).

The CLD was validated against primary data collected through semi-structured interviews held with experts working within, and conducting research on, food systems and food packaging in Australia. The validation process establishes confidence and usefulness in the CLD (Senge and

Forrester, 1980), by confronting and refining the model with data and expert opinions (Stermann, 2000). Validation in system dynamics is not the final step of the methods, but rather an important component that reinforces the process (Luna-Reyes et al., 2012). There is no single test that 'validates' a system dynamics model. Instead, confidence increases as the model passes more tests (Senge and Forrester, 1980). The main purpose of the CLD validation process is to increase the general understanding and the current knowledge about the drivers of packaging use in the food system. The scope of the validation process is to understand the dynamic complexity and not to represent detail complexity (Hovmand, 2014a).

2.2. Expert selection

Individuals selected for semi-structured interviews were domain experts that have worked, or are working in, organisations, institutions and businesses that manage food systems or that aim to reduce the impacts of food packaging. Experts have a high degree of competence in the domain of inquiry (Guest et al., 2006) and are expected to have a more elaborate understanding of the elements, structures and behaviour of the system components and their interrelationships compared to people that do not work directly in food systems or in tackling food packaging challenges (Hmelo-Silver and Pfeffer, 2004).

Experts were selected strategically from a population covering academia, government, packaging manufacturing businesses, supermarkets and grocery stores, transnational corporations, umbrella organisations and NGOs, and industry (e.g. food packaging consultants, food producers and supply chain stakeholders). This strategic selection requires some prior knowledge of the experts and is called judgement sampling (Harrell and Bradley, 2009). In this research, it was less important that the expert selection was statistically representative, because their value is derived from their mental models and they can be considered opinion leaders within their category (Hovmand, 2014b). The experts were contacted via email, website contacts forms and on some social media platforms. A mix of these contact channels was used for follow-up messages.

In qualitative research, advice from the literature on the number of participants to include in a study differs substantially (Saunders and Townsend, 2016). In this study the principle of saturation was used to limit and justify the number of expert participants (Morse, 1994; Guest et al., 2006). Theoretical saturation is reached when no additional data are found (Guest et al., 2006), which corresponds to the moment at which the involvement of additional participants produces little or no new themes or variables. To establish a transparent saturation protocol, the same set of questions was asked of all experts (Guest et al., 2006). Since we were sampling from a heterogeneous group we expected that a relatively large number of participants would be required to reach saturation (e.g. 12 to 50) (Saunders and Townsend, 2016). The final number of participants was also influenced by time, resources and the willingness of experts to participate.

2.3. Semi-structured interviews

Expert interviews were designed to explore the dynamic complexity of food systems using a set of *how* and *why* questions (Leonard-Barton, 1990). This enabled us to compare and contrast the variables and relationships influencing food packing use identified by experts, with those identified in the literature review (Chakori et al., 2021). The questions are provided in the supplementary materials (Table A7). The purpose of this process is to refine the literature-based model using expert knowledge.

In a semi-structured interview, a participant has time to describe their understanding of the factors influencing the problem (the use of food packaging), causality and feedbacks (Luna-Reyes et al., 2012). Several studies have used semi-structured interviews with experts to build and validate Causal Loop Diagrams (Aikenhead et al., 2015;

Belayutham et al., 2016; Peraphan et al., 2017; Agnew et al., 2018; Littlejohns et al., 2018; Mills et al., 2021; Xia et al., 2021).

In addition to answering semi-structured questions, experts were asked to complete a connecting circle, which is a systems thinking tool (Mildeová, 2013). The connecting circle allowed experts to explicitly explain interconnections between variables that they mentioned during the interview. The results from each semi-structured interview were merged into a single CLD, hence the results from each interview became additive (Eker and Zimmermann, 2016).

The interviews were conducted between March and August 2020 with each participant interviewed individually. Due to the COVID-19 pandemic, interviews were conducted online using Zoom, a video communication platform. Video calls can be a substitute for face-to-face in person interviews (Luna-Reyes et al., 2012). All interviews were audio-recorded and transcribed.

Content analysis was conducted on the semi-structured interview transcripts using NVivo (version 12.6.0) coding to identify variables and relationships (Eker and Zimmermann, 2016). To check for data saturation, the results of interview analysis were reviewed after each set of 6 interviews.

2.4. Network analysis

In addition to the interviews, a network analysis was undertaken using Cytoscape, software used for complex data network visualisation and analysis (Scardoni and Laudanna, 2012). This was to test the CLD and see if it generated a structure that corresponded to complex real-world systems. An appealing feature of Cytoscape software is that it can rapidly compute a set of accepted topological parameters from the input data. Finding a scale-free property in the system reinforces the confidence in the system built. Many real-world networks are not random and converge to similar structures (Barabási, 2009).

In network analysis the centrality test provided additional information on which variables in the systems are the most 'connected' and 'influential'. While this is not a systems dynamic modelling step, the combination of these two methods enhances the analysis of the complex system. A three column (factor A, factor B, polarity) network file and a two columns node file (factor, total connections) were generated in Microsoft Excel. These files were sequentially imported into Cytoscape software version 3.7.2 (Shannon et al., 2003). The topology of the network was first visualised using the Prefuse Force Layout clustering algorithm and every node awarded a suite of topological parameters using the Analyse Network function. The network was mapped using degree to map node size and polarity to map edge colour. The topological parameters were then exported and deployed for further analysis. Additionally, the radiality, which is a node (variable) centrality index, has been adopted to analyse the system (Scardoni and Laudanna, 2012). The radiality measure provides information about each variable's influential role in the system (connectedness and reachability), which refers to the degree a variable's relations reach out into the system providing access to many and diverse other variables (Valente and Foreman, 1998). A high radiality value translates into a variable that is generally closer to the other nodes and that has a high influence on the system, whereas a peripheral variable would have a low radiality value (Scardoni and Laudanna, 2012).

3. Results

3.1. Expert interviews and data saturation

We interviewed experts from academia, government, industry, non-governmental organisations (NGOs) and umbrella organisations, packaging manufacturers and retailers. The interviews lasted between 33 and 74 min each. A total of 223 messages (via email, website contact forms and on some social media platforms) were sent to 122 experts. Table A1 (supplementary material) presents the number of experts contacted and

interviewed for each sector. Data saturation occurred after the analysis of the 18th interview, with 27 variables identified in total and no new CLD variables emerging. Table A2 (supplementary material) shows the code definition changes by round of analysis. Supermarkets represent the most contacted category, yet they are the smallest represented category, with just one participant. The NGOs were the most represented category, with eight participants.

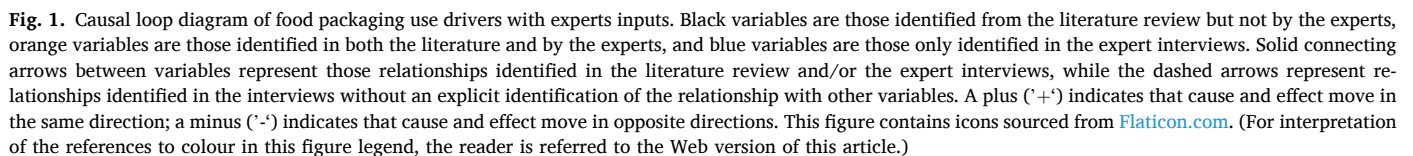
3.2. Drivers of food packaging use

The developed CLD of the food packaging drivers is shown in Fig. 1. This CLD is presented as a 'rich picture', a medium that captures the main entities and their crucial relationships and that invites a structured discussion to question the reality (Checkland and Poulter, 2020). This CLD is shown as an overlapping 'subsystem diagram' (Sterman, 2000) that groups variables together under four themes. Three of these subsystems (*globalisation*, *supermarkets* and *households*) show the drivers of food packaging. Conversely, the *environmental and health impacts* subsystem only shows the effects of food packaging. The supplementary material contains a more complete CLD (Figure A1).

The CLD contains four balancing feedback loops and 22 reinforcing feedback loops. These feedback loops are summarised in the supplementary material (Table A3). The CLD of the food packaging system is colour-coded to show the provenance of its variables and their interconnections. The colour-coding compares and contrasts the variables that were previously identified through a literature review (Chakori et al., 2021) with those identified through the expert elicitation interviews. All the experts have some understanding of some variables of the subsystems, but none of the participants had a full and clear understanding of the drivers and their connections within the subsystems. This highlights the importance of developing a qualitative model in the form of a CLD which synthesises data from the literature and expert interviews.

The drivers of food packaging identified through the expert interviews were grouped by subsystem categories and presented in Fig. 2. As presented in the CLD, some variables belong to more than one subsystem. Additional information on the 27 variables and their hierarchical categorisation are presented in Table A4 (supplementary material). The transcripts often included sentences that directly referred to the drivers (variables), but they also contained statements that referred indirectly to the variables. Participant responses have been summarised in the CLD with assigned variable names (Fig. 1), but Fig. 2 presents a more specific and rich code name list as used in NVivo for the transcript analysis. Examples of participant statements are provided in Table A5 (supplementary material).

Expert responses to the first question (see questions, Table A7, supplementary material) were focused on 'food safety', 'food waste', and 'technology' as the primary drivers of food packaging use. However, the interview process encouraged participants to expand on these concepts. For example, food safety (mentioned in 13 out of 18 transcripts) was connected to 'longer supply chains' by nine participants. 'Convenience' was the most frequently identified variable (mentioned in 16 out of 18 transcripts) and identified by all six groups, followed by 'corporation size and power' and 'supermarkets' model' (both mentioned in 13 out of 18 transcripts). Generally, there is a clear understanding of the role that packaging plays in increasing profitability of retailers and food corporations in the globalisation and supermarket subsystems, given experts identified all the variables belonging to loop R1 and R4. The household subsystem has the greatest number of variables. Participants identified several variables that belong to this subsystem, however, concepts such as 'convenience', 'pre-prepared food' and 'take away food' have similar meaning. In order to simplify the CLD these terms have been integrated into the variable 'convenience ready-to-eat meals'.



The network analysis creates a visualisation that differs from a CLD. While the CLD provides a road map of cause-effect relationships, the network analysis shows the relative level of influence that each variable is predicted to have on signal transmission across the entire network (Borgatti et al., 2009). The Cytoscape visualisation produced can be considered an entirely data-driven mathematically formal structure. The model resulted in a network with a scale-free connectivity distribution, in line with many real-world systems (Barabási, 2009) (see graph A2, supplementary material).

This analysis also focused on the radiality measurement. This awards high values to nodes considered influential because of a low average shortest path length. Network theory predicts these high radiality nodes

The variable 'attractiveness to time pressed consumers' has the highest radiality coefficient (Fig. 4) and also a high (6) number of connections in the CLD (Fig. 1). The second most influential variable is 'convenience ready-to-eat meals' (with 7 connections). The radiality ranking (Fig. 4) reflects the importance of 'convenience', which was also the most frequently identified concept in the interviews (Fig. 2). Despite the low number of connections (3), 'sales revenue' is positioned as 4th variable with the highest radiality coefficient, meaning that it possesses an influential role in the network. The variable 'packaged food sales' is positioned as 6th in the radiality ranking, although it is centrally placed in the CLD. Food packaging is just a symptom of a system driven by other, more influential, variables that lead to the need for food packaging use. The top 10 highly influential variables belong to the globalisation and household subsystems. The supermarket subsystem could be considered an intermediate system influenced mainly by the needs of the global market and by household dynamics. In conclusion, the CLD built from the literature, the interview responses, and the network analysis provides rich information on the behaviour of the system and on the drivers of food packaging use.

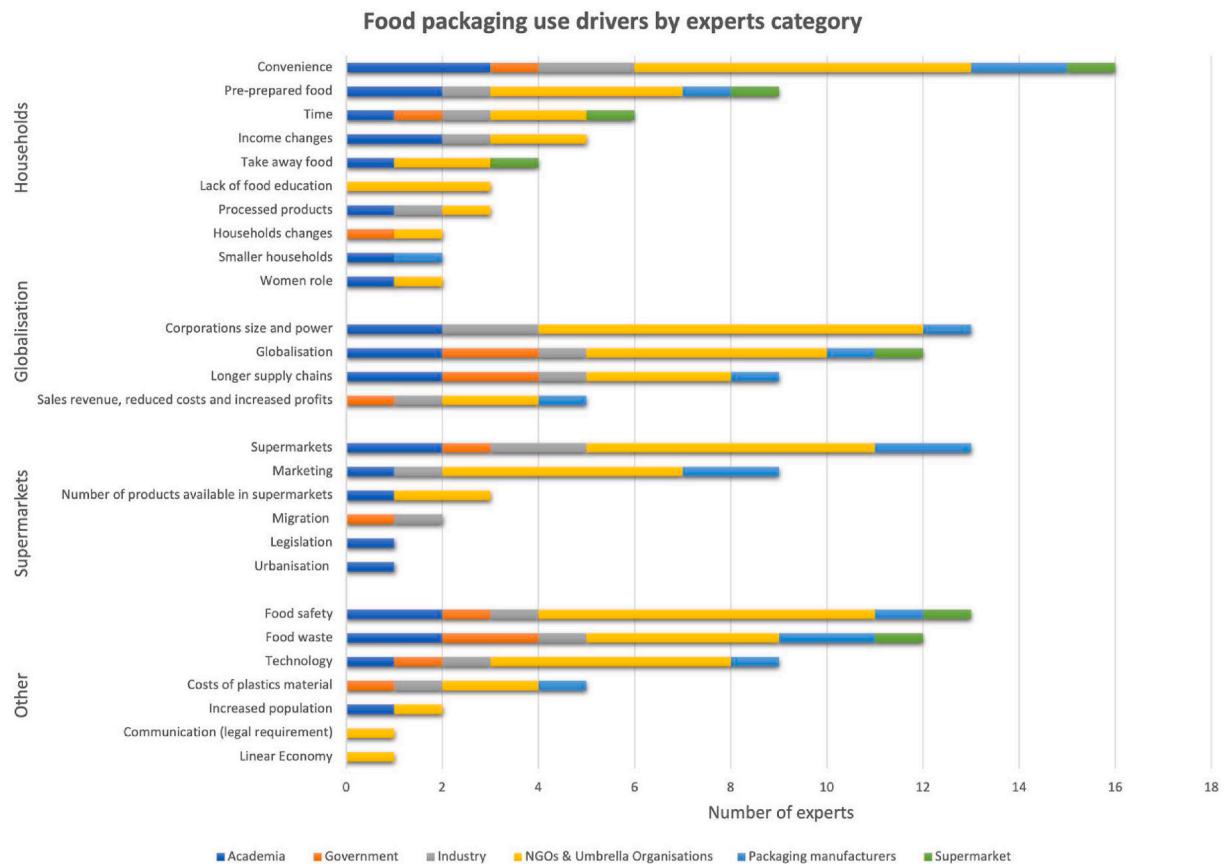


Fig. 2. Drivers of food packaging use resulting from expert semi-structured interviews.

3.4. Interventions as per participants' knowledge

In addition to identifying drivers of food packaging use, the experts were asked to identify and suggest interventions to reduce and manage food packaging issues. Table 1 presents the results showing six categories of intervention. Recycling (14 experts), legislation (11 experts) and new sustainable material (10 experts) were the most common interventions identified.

4. Discussion

4.1. Food packaging drivers versus current interventions

Our research illustrates that food packaging is a wicked problem, rooted in a complex food system with numerous interlinkages and feedback loops that transcend areas of individual knowledge and subsystems. For example, reinforcing feedback loops R5, R14, R15, R17 (Fig. 1, Table A3) encompass all three driver subsystems. The presence of these feedback loops means that food packaging problems cannot be reduced to simple and independent cause-effect relationships. Moreover, the process of building the CLD sheds a light on stakeholder mental models about food packaging use. Mental models give us information on how experts think the dynamic system works (Luna-Reyes and Andersen, 2003). Our findings show that the knowledge of key actors within the system does not encapsulate the whole system. This is exemplified by expert knowledge of globalisation's dynamics and a less clear understanding about the other two driver subsystems. However, we acknowledge that participants had limited time that might have influenced the lack of reflection from a whole-of-system perspective.

The data elicited from the experts in this study showed that even if the experts explicitly identified drivers of food packaging-use, the solutions proposed by them (presented in Table 1) do not tackle the

underlying drivers. Rather, their identified interventions are indicative of a 'shifting the burden' archetype. This system archetype (Senge, 1990), which is shown in Fig. 5, represents a situation where symptomatic solutions are favoured over fundamental solutions in addressing a problem. Symptomatic solutions are typically favoured because they are easier to implement and have a faster response (Senge, 1990). However, the outcome of this is the emergence of some unintended consequence or side effect that causes the problem to return, while at the same time making it harder to fix using the fundamental solution (Kim and Anderson, 1998).

Traditionally, studies on food packaging have taken a more siloed approach, focusing on the consequences of post-consumer packaging management and on the introduction of new materials (Lebreton and Andradóttir, 2019; Nemat et al., 2020; Chakori et al., 2021). Solutions that aim at introducing new 'sustainable' packaging materials in the market do not question their disposability, or how and why these materials enter the economy (Lucas, 2002). For example, 'sustainable' packaging made of biomass-based materials is still designed to be single-use, with impacts on resource-use that have detrimental effects on ecosystems and food production for human needs (Chakori et al., 2021). Recycling schemes, or the introduction of legislation to ban some plastic items from the market, are also symptomatic solutions. Symptomatic solutions create side-effects that contribute to the erosion of a more radical solution (Kim and Anderson, 1998). For example, increased efficiency in recycling practices can lead to the occurrence of the Jevons Paradox (Figge et al., 2014; Korhonen et al., 2018), in which initial environmental gains are ultimately offset by higher biophysical throughput due to increased production and consumption of 'green' products. Moreover, the health's risks of recycling packaging materials into new food packaging, or using new types of food packaging materials, need to be assessed and avoided. Chemical safety consideration (i.e. increases in sources contamination and chemical migration levels) must be

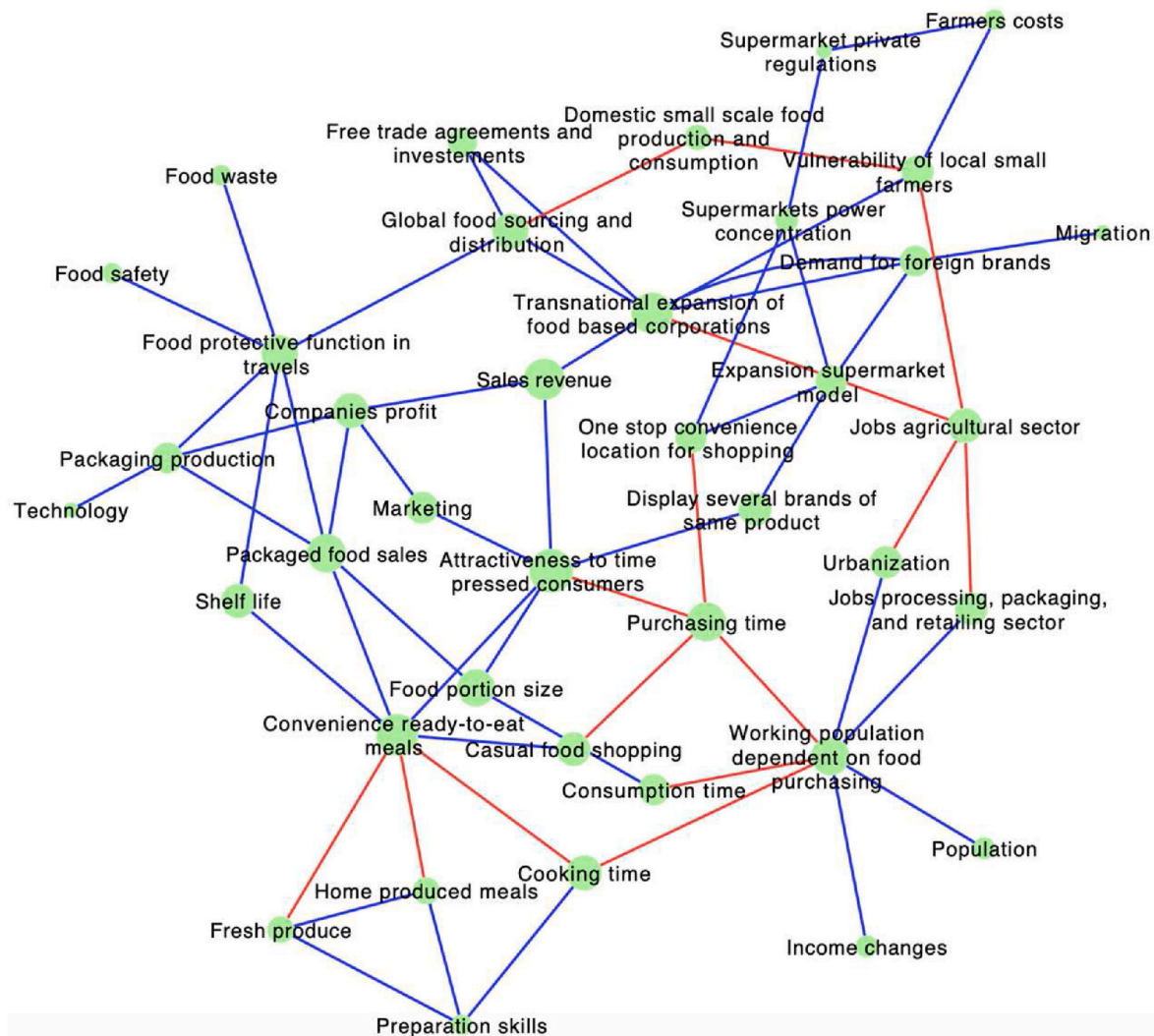


Fig. 3. Network Map. Node size has been mapped to degree (number of connections), edge colour has been mapped to polarity: blue for positive connections and red for negative ones, following the same polarities of the CLD. The overall structure has been visualised using the Prefuse Force Directed Layout which clusters ‘modules’ of highly inter-connected nodes close together, and places nodes with high radiality (overall network influence) in the centre of the network. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

prioritised, alongside environmental benefits, to avoid regrettable substitutions (Geueke et al., 2018; Muncke et al., 2020).

A clear gap remains between the existing interventions that focus on the impacts of food packaging materials (e.g. waste production), and the need for interventions that address the underpinning drivers of packaging use. When asked to reflect on and explain the gap between the drivers and the proposed interventions, some experts said that “the [Australian] government’s first strategy is waste avoidance. And from a retailer and manufacturer perspective, convenience is become so significant, that they won’t entertain that” (participant 17). Another participant explicitly mentioned that tackling the drivers of food packaging use would go “completely against the market economy; against the growth model and marketing economy” (participant 6), with another participant stating that “it’s seen as too risky for governments to even talk about [redesigning food systems] because that would threaten businesses and employment, as well as consumer choices, and it’s part of the antagonism within some groups to the environment movement” (participant 2).

The predominant focus on indicators of impact without consideration of the determinants of the problem are indicative of how wicked problems are typically managed from a non-systems perspective. Systems theory posits that impacts are ‘indicators’ that reflect the behaviour of a complex system and do not govern its behaviour (Senge, 1990).

Rather, it is how the variables are interconnected that give rise to the behaviour (impacts) of a system and therefore meaningful intervention needs to be informed by the system structure. The risk of these symptomatic solutions, also referred to as greenwashing practices (Khmar and Kronenberg, 2018), is that they lessen the pressure to implement a more systematic fundamental solution that requires commitment to change, along with more time and energy (Kim and Anderson, 1998).

4.2. From symptomatic solutions to a fundamental solution: the shift towards a degrowth economy

The radiality analysis (Fig. 4) highlighted the most influential drivers of the system (e.g. ‘attractiveness to time pressed consumers’, ‘convenience ready-to eat meals’, ‘transnational expansion of food-based corporations’, ‘sales revenue’, ‘global food sourcing and distribution’). These drivers, and consequently the three CLD driver subsystems, could be considered as pillars forming the structure of the industrial, capitalist and globalised growth-driven food system. Globalisation enables an increase in transnational corporations’ profits (loops R1, R16, R20) (Folke et al., 2019; Chakori et al., 2021) and the power held by these corporations and supermarkets reflects what in system terms are called ‘success to the successful’ feedback loops, which are positive feedback

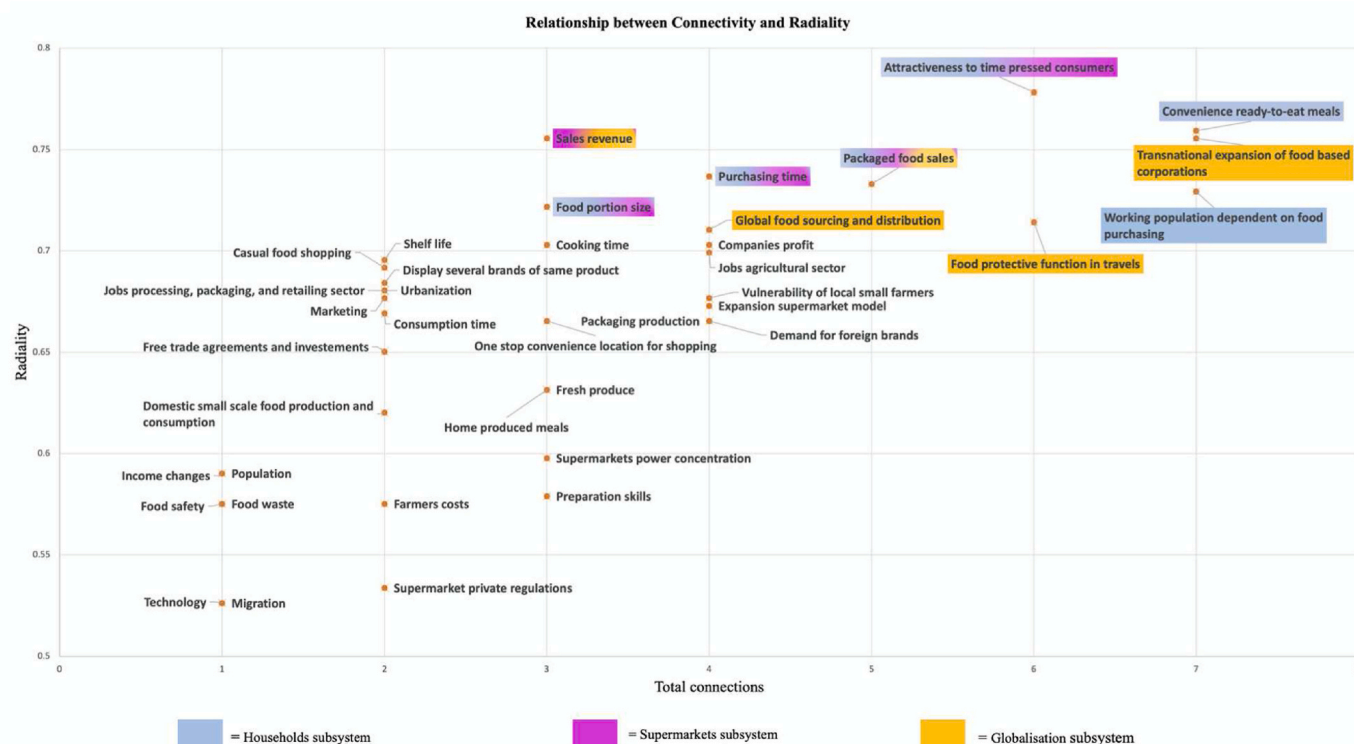


Fig. 4. Relationship between connectivity and radiality. Radiality is a function of connectivity. The 10 most influential variables have been coloured in relationship to the subsystem to which they belong. If a variable belongs to multiple subsystems, multiple colours are displayed in the variable's name. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1

NVivo Nodes: Food packaging current interventions.

Node name & Transcript Summary	No. expert
Recycling	14
- Use of recycled and recyclable materials	
- Improvement of recycling industry (e.g. chemical recycling)	
Legislation	11
- Mandate and implement stricter recycling schemes	
- Ban export of waste	
- Ban some items from the market (e.g. straws, plastic bags)	
- Introduce sustainable packaging guidelines (e.g. compostable standards)	
- Product stewardship and extended producer responsibility frameworks	
New 'sustainable' materials	10
- Biodegradable and compostable materials	
- Alternative plastics (e.g. starch-based; bamboo fibres)	
- More technological innovation	
- Increase access to compostable waste collection points	
- Make "sustainable" options more convenient for consumers	
Reuse and refill	9
- Introduce and implement "returnable" packaging (e.g. loop systems)	
- Bulk buying and refill models	
Industry initiatives	6
- Voluntary recycling	
- 'Technological innovation' for end-of-life material recovery	
- Reduce unnecessary packaging in volume	
Consumer education	5
- Educate on recycling options	

loops that reward the successful with the means to succeed (Meadows et al., 2004). Reinforcing feedback loops magnify change, leading to self-perpetuating behaviour that can be classified as vicious (but also virtuous) cycles (Sterman, 2000). Meadows et al. (2004) highlight how these structures tend to be endemic in a society that does not implement counterbalancing structures. The capitalist economic model is fuelled by

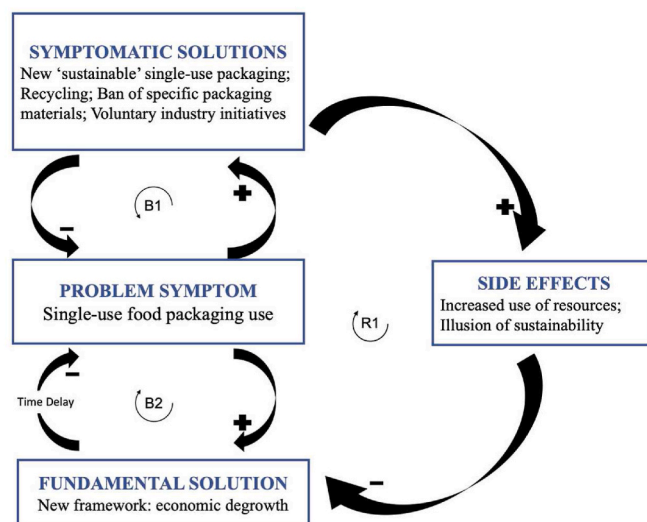


Fig. 5. Shifting the burden archetype. A plus ('+') indicates that cause and effect move in the same direction; a minus ('-') indicates that cause and effect move in opposite directions.

the continuous accumulation of capital through the production of commodities (Fine, 1994; Blauwhof, 2012; Kallis et al., 2012). Food systems are not immune to this economic imperative. Ioris (2018, p. 4) points out how there has been a "movement from food-as-nutrition and agriculture-as-social-integration to a situation in which agri-food operations are carried out primarily to circulate and accumulate capital. In other words, the role of agriculture as a source of nutrition and livelihood is being increasingly supplanted by the imperatives of money and profit, which is happening in the wider context of a globalized

market-based society in which everything is prey to commodification". The process of displacement of production for use by production for market's profit is defined as commoditisation (Fine, 1994). Many experts in this study (12 out of 18) mentioned food packaging as a tool to reduce food waste, indeed many studies highlighted the importance of packaging in prolonging shelf-life and reducing spoilage (Oki and Sasaki, 2000; Marsh and Bugusu, 2007; Gustavo et al., 2018). These food packaging functions have been represented in the CLD with the icons "shelf life" and "food protective function in travels". Nevertheless, it would be reductionist to consider that the absence or presence of packaging is the main factor influencing food waste. The fundamental question that should arise is why our food systems generate food waste? The response to this question is beyond the scope of this study, the generation of food waste should be considered separately, aiming at understanding why the problem exists and persists. Studies point out at various factors that contribute to food loss (suppliers' side) and food waste (retailers and consumer side), such as actors' decisions in the food supply chain to overproduce food in order to maximise profit, market prices influence, retailers' rejection based on food appearances (private quality standards), consumers opportunity costs and attitudes (Devin and Richards, 2018; FAOFAOotUN, 2019; Messner et al., 2020).

Under the growth-driven model, food businesses (e.g. agribusinesses, packaging manufacturers, the oil and gas industry) are governed by the imperative of profitability and not by the goal to satisfy food needs in a socially just and ecologically sustainable manner. Food preparation, preservation, and packaging serves the expansion of the international food trade and the centralisation of food retailers, while responding to the demand for convenience by time-poor consumers. An economic system based on the profitability of import and export markets translates into a food system that relies heavily on food packaging for protection and transport. Moreover, the increased trade with distant markets benefits countries economic 'success' by increasing the Gross Domestic Product (GDP), which remains the main economic index used to measure economic growth. This index ignores the social and environmental costs of the growth-driven system (Costanza et al., 2014). In the current system, social and environmental negative effects (costs) are externalised, not included in the real cost of products and not compensated by the parties producing them (Eidelwein et al., 2018). The growth-driven food system is driving the increase in packaging use. Under the current growth-driven economy, 'sustainable' packaging solutions would still be used for market expansion.

As presented in the previous section, the growth-driven economy is responsible for the food packaging addiction of the modern food system, therefore, a shift towards an alternative economic model should be considered. The implementation of concepts such as the circular economy, cleaner production, green businesses, green growth is not contributing towards systemic change (Wiedmann et al., 2020; Grafström and Aasma, 2021; Sundqvist-Andberg and Åkerman, 2021), hence here we consider economic degrowth as a new system to be considered. There is a growing literature exploring economic degrowth (Schneider et al., 2010; Asara et al., 2015; Gomiero, 2018; D'Alessandro et al., 2020; Wiedmann et al., 2020), which is an economic framework that aims at "an equitable downscaling of production and consumption that increases human well-being and enhances ecological conditions at the local and global level, in the short and long-term" (Schneider et al., 2010, p. 512). From a shifting the burden archetype perspective, economic degrowth could be framed as a fundamental solution (Fig. 5). Systemic change is defined as a fundamental change in the way the system is structured (Polhill et al., 2016). Degrowth focuses on voluntary and fair processes of constraining production and consumption to reduce the ecological footprint of the global economy (Khmara and Kronenberg, 2018). To transition towards a degrowth framework, with the goal being reduced food packaging use and increased food systems sustainability, policies should, for example, aim at re-localising and decentralising the economy, enhancing local food production and reducing international food trade. The localisation of food systems

should not be understood simply as the localisation of food production and consumption in spatial relations, as this would not be enough to reduce food packaging and increase the general sustainability of food systems. Localisation must be more critically analysed. It is important not to fall into the 'local food trap', in which local food production is promoted as the most sustainable option as an ideology to counter globalisation (Hinrichs, 2003). Localisation does not automatically translate into social and ecological improvements. Re-localisation should be intended as a shift away from the commoditisation of food systems and towards the emergence and revitalisation of new social relations that foster the interest of local eaters and food producers, while respecting environmental sustainability. Local-scale economies are more manageable (Ajulo et al., 2020) and shorter supply chains would curb the need for packaging, e.g. stimulating systems that encourage reusable containers. Shorter supply chains would translate into shorter food shelf life and a decreased need of food protection. Shorter supply chains would decrease the need for primary, secondary and tertiary packaging. Additionally, by consuming fresh local produce, food packaging used for globally processed and ultra-processed food could be minimised. This study proposes shorter supply chains as a packaging avoidance strategy to reduce the packaging addiction of the current global food system, but it acknowledges that also a more localised food economy would require some level of food packaging and containers.

A certain level of global food exchange that aims at responding to food security and other basic needs would remain necessary, but the growth-driven globalisation needs to be downscaled. Localisation should also have attributes of decentralised food systems to counter-balance the increased power of supermarkets. Sustainable alternatives exist: farmers markets (Ajulo et al., 2020), Community Supported Agriculture communities (Bloemmen et al., 2015), and all the social enterprises that satisfy real needs de-emphasising the importance of profit and without externalizing social and ecological costs (Johanisova et al., 2013). These examples can be considered shadow networks, which are networks that operate outside the dominant system and function as incubators for new practices (Boonstra and Joosse, 2013). Other studies explore in details the various social and ecological steps that could be undertaken in order to improve the general sustainability of food systems while guaranteeing people a proper supply of food, in terms of quality and quantity (Chaudhary et al., 2018; Gomiero, 2018; Poore and Nemecek, 2018). However, alternative food systems still face structural constraints, as legislative rules prioritise agribusinesses (Gerber, 2020). Wiedmann et al. (2020, p. 4), pointing out to political barriers and to the capitalist class, write "there is a common interest in maintaining the capitalist system and favourable conditions for capital accumulation". Hence, degrowth remains rarely considered in the context of business activity, because of the conflict between its focus to downscale the economy (in rich countries) and the role of businesses in GDP growth (Khmara and Kronenberg, 2018).

As packaging serves transnational corporations' profit accumulation, by allowing the expansion of the globalised market, intensive marketing campaigns and competition on supermarket shelves (Chakori et al., 2021), considering the food economy as not-for-profit would change the dynamics of the subsystems presented in the CLD (e.g. loops R3, R7, R10, R11, R14). As food is a fundamental necessity for human (and non-human) life, could we envisage a not-for-profit food economy? Not-for-profit businesses have social and environmental missions and use all the surplus for social benefit (Hinton, 2020). Using a series of Causal Loop Diagrams, a recent study of Hinton (2020) explored not-for-profit business structures in relation to a transition towards a degrowth society. The not-for-profit model would contrast with current agri-food operations that are carried out primarily to circulate and accumulate capital (Ioris, 2018). Ioris (2018) emphasises the importance to reinstate food as food, farmers as food producers and markets as allocation tools. Not-for-profit food systems might enable these conditions. Not-for-profit food systems would need to be measured with indexes that go beyond GDP and that internalise all the true costs of

products. How we evaluate food systems matters and it influences their structure. Alternative measures of progress, that focus on social and ecological well-being factors exist (Van den Bergh, 2010; Costanza et al., 2014) and should be adopted to develop a more sustainable food economy. The not-for-profit food economy would discourage the profit-driven nature of transnational food corporations, alleviating the environmental and social pressure caused by the continuous expansion of production and consumption. The power held by transnational corporations create barriers to smaller firms (loop R10 and R11). In a not-for-profit food economy, smaller and local food systems might face less barriers in their establishment as profit maximisation and GDP growth would not be prioritised by governments and the private sector. Re-localising food circuits would reduce transport and the need for packaging, but localisation must be accompanied with other economic degrowth transformations that would also change the household dynamics presented in the CLD.

Both the household subsystem in the CLD and the radiality analysis show that the need for convenience is driven by time pressure experienced by consumers. Convenience seems to be just assumed as a normal way of living without questioning the need for it, in particular in regard to food production and consumption. Time poverty is a growing phenomenon in modern societies, with people needing to manage paid labour, unpaid labour (e.g. family duties) and personal care (e.g. eating, sleeping) (Goodin et al., 2005). Time poverty impacts our lifestyle and diet (Giurge et al., 2020), resulting in an increased consumption of processed and ultra-processed packaged food, which also contributes to health issues, such as non-communicable diseases (e.g. cardiovascular and cerebrovascular diseases) (Banwell et al., 2005; Srouf et al., 2019; FAOFAOotUN, 2020). Overweight and obesity conditions, that result from processed food consumption, increase the risks of experiencing non-communicable diseases, and remain a pervasive and growing trend (FAOFAOotUN, 2020; Rockström et al., 2020; Moradi et al., 2021). Furthermore, time poverty and a certain level of environmental consciousness might risk increasing processed food consumption. With the awareness of the unsustainability of our food systems, in particular related to meat consumption, there is an increasing market providing processed packaged vegetarian products (e.g. plant-based protein burgers) (Detzel et al., 2021). The consumption of these products might exacerbate the ecological and health issues related to packaged ultra-processed food, especially if, like Detzel et al. (2021, p. 11) have reported, these products ultimately serve market expansion (i.e. Jevons Paradox). Analysing this aspect of food systems goes beyond the scope of the study, however, it is important to highlight that a shift towards a vegetarian diet, especially in Global North countries, could be undertaken by consuming non-processed or minimally processed, unpackaged or minimally packaged, locally produced vegetables, fruits, whole grains, legumes (Kahleova and Katz, 2020). As Kahleova and Katz (2020, p. 1) write, “vegetarianism involves not only the avoidance of meat, but a general elevation of diet quality”. Moreover, the combination of time poverty and ultra-processed packaged food consumption also affects social relationships (Monteiro et al., 2018). Convenient ready-to-consume meals displace shared experiences of acquiring, preparing and cooking food, individuals tend to eat separately and at different times, weakening the social life of households (Monteiro et al., 2018).

Therefore, time-poverty should be addressed with substantial systemic change to increase time availability, and in particular mealtimes. By liberating people from the constraints of the labour market, policies could reduce the tension between cooking and working time. The economic degrowth literature suggests different ways to organise work, stressing the importance to increase (free) available time by, for example, reducing working hours and implementing work-sharing mechanisms, or by adding a universal basic income and income ceilings (Kallis et al., 2012; Khmara and Kronenberg, 2018; Scarrow, 2018; Dula et al., 2019). People would be able to grow, cook and consume more unpackaged fresh produce issued from responsible agroecological

practices, reducing the need for pre-prepared packaged meals. Shifting society towards economic degrowth principles is a fundamental, much needed, solution that would influence food system sustainability.

Finally, across centuries, different types of food containers and packaging (e.g. clay pots, leaves, animal skins, glass, plastics) have been used to preserve and carry food. Various studies have presented the history and usefulness of packaging (Brody et al., 2008; Emblem, 2012; Paine and Paine, 2012; Baker et al., 2020). Packaging has brought several societal benefits; however, despite this, the addition of the global food system to packaging continues to cause detrimental environmental and health impacts (Pongrácz, 2007; Geyer et al., 2017; Muncke et al., 2020). This study explores economic degrowth as a new multi-dimensional paradigm able to shift society towards packaging-avoidance food systems by prioritising a critical relocation of food supply chains, an exploration of a not-for-profit food economy and an increased time-availability that modern citizens have. Although economic degrowth concepts are widely explored in the literature, further work is required to operationalise this sustainability transition (Khmara and Kronenberg, 2020). By operationalising degrowth principles, the shortcoming of this system can be analysed. As Khmara and Kronenberg (2020, p. 1) state, degrowth, as the sustainability transition domain, is simultaneously a “research area and the process it studies”. Hence, more research and implementation of degrowth models in practice are flaws to this can be discovered and studies.”

4.3. Systems thinking and network analysis: strengths and limitations

In research, triangulation occurs with the use of multiple sources of evidence, such as secondary data, direct investigation and theories (Lu and Sexton, 2009). To enhance confidence in findings, triangulation can also occur using different methods. Triangulation is particularly important in order to reduce any bias (Lu and Sexton, 2009). In systems thinking, interviews with participants could lead to contradictory information (Ryan et al., 2019), for example on the understanding of the causal relationship between variables. In our study participants responses converged to a similar understanding of the use of food packaging as a response of globalisation, supermarket power concentration and need for convenience, which coincided also with the trends of the behaviour over time found in the literature (Chakori et al., 2021). Moreover, the network analysis confirmed that the model built has a scale-free nature, consistent with other real-world systems (Barabási, 2009), which is an information not included in CLDs.

Several studies provide a clear explanation about CLD development and about data collection methods, however, it is less common to find descriptions of CLD analysis methods (Ryan et al., 2019). Thus, this study used a network analysis method to expand on the analysis of the CLD. Comparing and contrasting different models can highlight the unique characteristics about a phenomenon of interest (Ryan et al., 2019). System dynamics tools (CLDs and Stock and Flow models) provide information about the dynamics and behaviour of the system, while the network analysis is a static representation that does not give information about, for example, the reinforcing and balancing loops, as presented in the CLD. However, while CLDs are useful tools to understand the system behaviour and system's emergent properties, one of the limitations is the lack of information about the influence of each variable based on an understanding of network topology. For this reason, a network representation of the model has been adopted to facilitate the structural analysis of CLD and to support the measurement of the importance of the variables (Honti et al., 2019). The results of the network analysis provided information on the hierarchical importance of the subsystems, which enables us to design solutions that prioritise the elements of the system with the highest predicted influence.

Finally, the experts interviewed are based in Australia, but the CLD model could be used to understand the modern food system due to the globalised nature of the food economy. In fact, since the 1960s the food

system has been globalised and global north and global south countries embrace similar trends in food system changes, such as in the expansion of vertically integrated transnational corporations and for appearance and growth of the supermarket model (Chakori et al., 2021). Therefore, this study could be used as a lens and starting point to analyse other modern food systems.

5. Conclusion

The results of this study show that i) expert knowledge is limited to some parts of the system; ii) current interventions and solutions identified by the participants of this study do not tackle the problem drivers, hence the problem persists; and iii) the most influential food packaging drivers belong to the globalisation and household subsystems. We contextualised our findings in relation to systems theory and the wider literature, discussing the need for a systemic change in order to reduce food packaging use. Only experts in Australia have been interviewed, however due to the global nature of the current food system, future studies could validate the CLD with experts of other countries.

Symptomatic solutions that erode the viability of a more fundamental systemic redesign of food systems should be avoided. A transformation requires to simultaneously deactivate the causal loops mechanisms supporting the current food economy and to activate those allowing a different system to be actualized. The modern food system continues to rely on the pursuit of economic growth, accelerating the exploitation of the planet's resources, alongside severe negative effects on human health. We discussed the urgency to transition towards an economic degrowth framework, in which social and ecological well-being would be prioritised. In a degrowth framework international trade would be reduced incentivising a re-localisation of food systems; the food economy could be not-for-profit shifting away from the commoditisation of the food systems; time poverty needs to be tackled by increasing food time availability, by reducing labour time via various mechanisms.

CRedit authorship contribution statement

Sabrina Chakori: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, (reviewers' responses), Visualization. **Russell Richards:** Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, (reviewers' responses), Supervision. **Carl Smith:** Conceptualization, Methodology, Software. **Nicholas J. Hudson:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, (reviewers' responses). **Ammar Abdul Aziz:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, (reviewers' responses), Resources, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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